

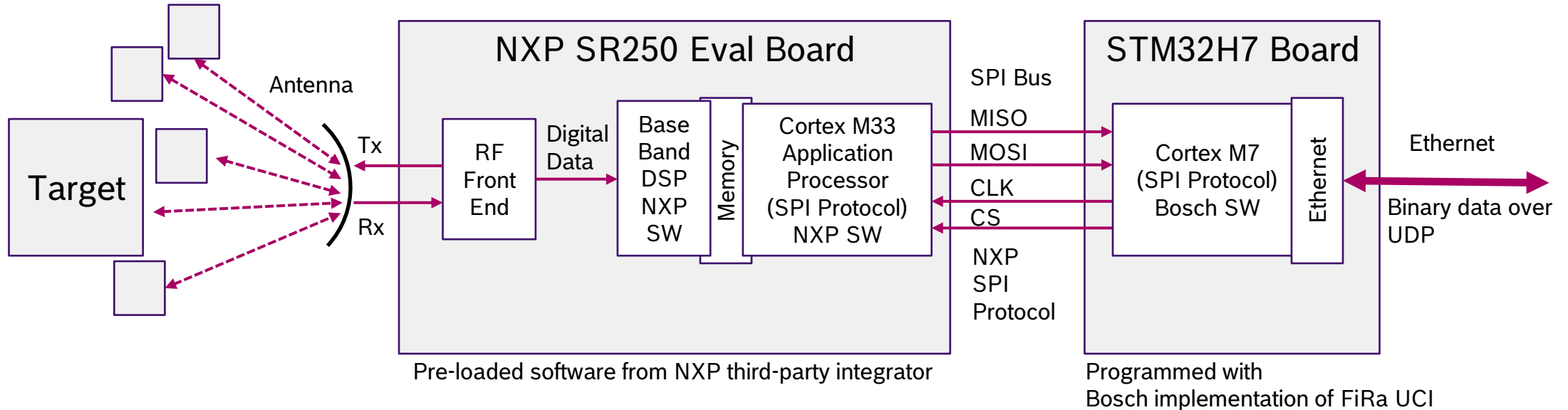


Bosch UWB Measurement Setup

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Hardware Setup Overview



FiRa: Fine Ranging Consortium [<https://www.firaconsortium.org/>]

UCI: UWB Command Interface

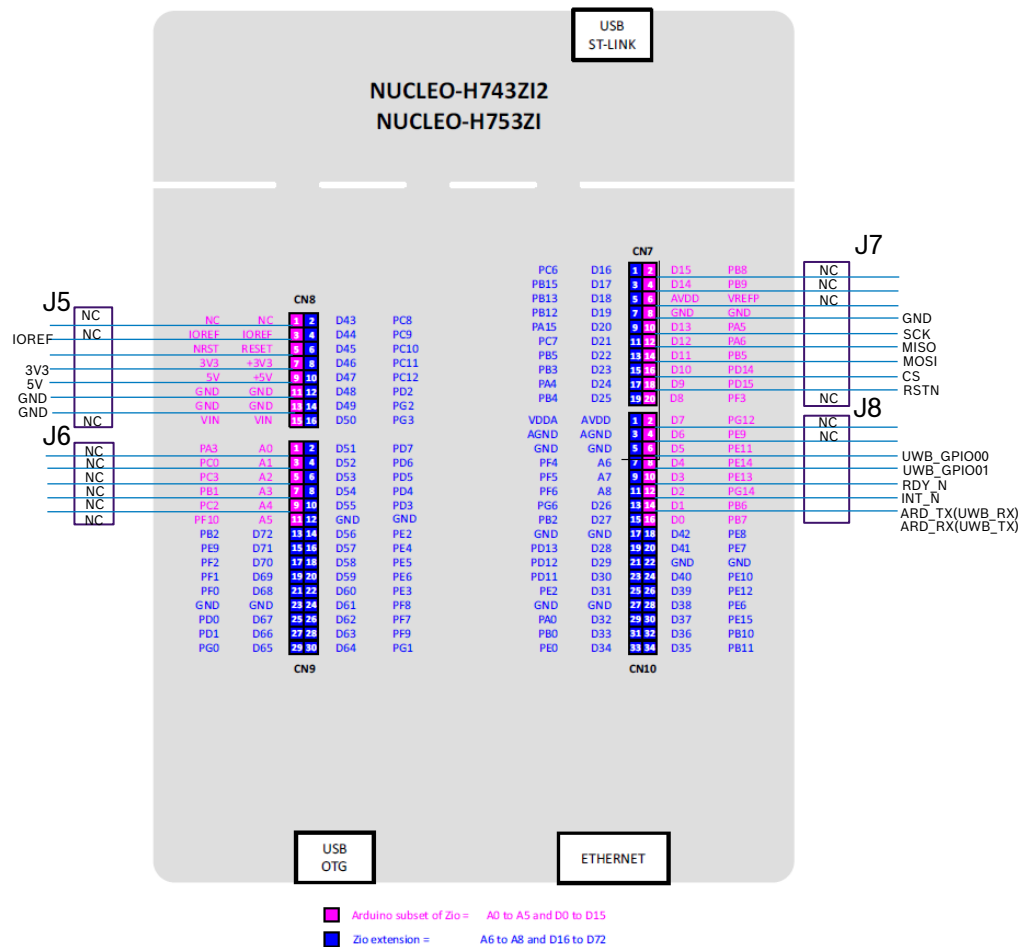
Internal C-SC1 | CR/RCS1-NA | 2026-02-27

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Pinout Maps

Figure 18. NUCLEO-H743ZI2 and NUCLEO-H753ZI



Nucleo	TS250
PA5	SCK
PA6	MISO
PB5	MOSI
PD14	CS
PD15	RST_N
PE11	UWB_GPIO00
PE14	UWB_GPIO01
PE13	RDY_N
PG14	INT_N
PB6	ARD_TX(UWB_RX)
PB7	ARD_RX(UWB_TX)

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Software and Documentation Pre-requisites

- Please install STM32CubeProgrammer software [\[Link\]](#)
 - Allows update of firmware on boards to add/modify/remove functions

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Ethernet Setup

- STM32H7 board is pre-configured to be on IP 192.168.1.10
 - Listening on port 37249
- Device connecting to STM32H7 over ethernet should be on IP 192.168.1.102
 - Listening on port 20000
- DHCP is best set to Manual on the device with IP 192.168.1.102

 Ethernet
Not connected 

Authentication settings

Edit

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Off 

Set a data limit to help control data usage on this network

IP assignment:

Manual

IPv4 address:

192.168.1.102

IPv4 mask:

255.255.255.0

Edit

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Ethernet Input Structure

```
typedef struct {  
    uint8_t cmd_id; // 1: Set radar configuration index and duration of run  
    uint8_t setting_idx;  
    uint32_t duration_in_ms;  
}udp_input_cmd_t;
```

Bosch UWB Measurement Setup

Ethernet Output Structure

[illegible]

0000	f4 2e 4e 20 10 12 00 00	01 00 00 f0 00 00 08 00
0010	80 00 05 00 00 00 21 f4	c4 00 13 08 32 00 e0 b8
0020	c4 00 01 61 01 01 08 05	02 04 21 3c 5b 00 00 00
0030	77 02 fd 00 f6 00 cc 07	2e 06 4b 0f 5b 0a 97 f5
0040	32 0c d7 c3 4f 31 66 d0	92 65 8f 15 bd 54 2e 2b
0050	5f 0e ef 06 0d f3 ee f5	8a 05 56 06 ea 08 42 0c
0060	35 f8 04 ff 90 f4 80 fa	2e fe 6b 03 d4 fe 9e 04
0070	0f f6 0c fc 1d f6 b6 fb	5c fd 94 00 ec ff cf fd
0080	18 f9 f4 fb 1f fe ca fe	47 00 e4 fe 4b ff b4 fe
0090	3b 00 91 00 19 00 1f 01	d1 fe 18 00 f6 fe dd ff
00a0	fd ff 65 00 19 00 73 00	4c ff d8 ff 0e ff 75 ff
00b0	f0 ff f9 ff 9c 00 9e 00	30 00 8a 00 d3 ff 70 00
00c0	44 00 bf 00 65 00 7b 00	bb ff e5 ff a6 ff e9 ff
00d0	21 00 08 00 41 00 01 00	2e 00 1b 00 05 00 fe ff
00e0	c9 ff bd ff ff ff d3 ff	5c 00 1f 00 71 00 34 00
00f0	34 00 12 00 09 00 e6 ff	0b 00 cd ff f7 ff c3 ff
0100	db ff b8 ff e0 ff c8 ff	06 00 f3 ff 09 00 f6 ff
0110	ea ff e0 ff e2 ff ea ff	ed ff 01 00 f8 ff 11 00
0120	fa ff 25 00 f8 ff 26 00	f5 ff 13 00 0e 00 1c 00
0130	24 00 14 00 03 00 12 00	f7 ff 0f 00 05 00 05 00
0140	02 00 00 00 03 00 f0 ff	03 00 ed ff 03 00 f1 ff
0150	0e 00 05 00 0b 00 0a 00	ff ff 0c 00 f8 ff 03 00
0160	fb ff 14 00 06 00 28 00	e6 ff 0a 00 d6 ff ea ff
0170	ee ff ec ff 03 00 f6 ff	02 00 f4 ff fd ff f9 ff
0180	0d 00 fa ff 01 00 f1 ff	01 00 fc ff 13 00 0e 00
0190	12 00 07 00 01 00 f4 ff	fc ff f3 ff 03 00 0e 00
01a0	05 00 11 00 f4 ff f7 ff	f0 ff f8 ff fd ff 01 00
01b0	02 00 03 00 ff ff fb ff	fc ff f5 ff 06 00 fc ff